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CURRICULUM VITAE OF CHINYERE GLORIA MBAKA

PhD Candidate, Electrical and Computer Engineering
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Education

Doctor of Philosophy in Electrical and Computer Engineering
Rochester Institute of Technology, Rochester, NY
August 2023 to Present
GPA: 3.62 out of 4.0

Relevant coursework includes Analytical Topics in Computer Engineering, Research Methods in Electrical and Computer Engineering, Interdisciplinary Research Methods, Applied Programming in C, Interface and Digital Electronics, Real Time and Embedded Systems, Machine Intelligence, Doctoral Seminar, and Human Factors in Artificial Intelligence.

Master of Technology in Electronics and Telecommunication Engineering
Obafemi Awolowo University, Ile Ife, Nigeria
2019 to 2021

Coursework included Measurements and Instrumentation, Digital Signal Processing, Control System Dynamics, Communication System Engineering, Antenna Theory, Satellite Communication Systems, and Mathematical Methods in Engineering.
Master’s thesis focused on duct propagation and its application to naval communication at sea.

Bachelor of Engineering in Electrical and Electronic Engineering
Federal University of Technology, Owerri, Nigeria
2002 to 2007
GPA: 3.99 out of 5.0

Undergraduate coursework included Semiconductor Devices Technology, Microprocessor and Microcomputer Systems, Control Systems Engineering, Data Communications, Reliability and Quality Assurance in Electronics, and Introduction to VLSI System Design.
Final year project involved the design and construction of a microcomputer based wireless security system.

Research Experience

Graduate Research Assistant
Respiratory Technologies Laboratory, Rochester Institute of Technology
August 2023 to Present

I contribute to the design, deployment, calibration, and performance evaluation of Wireless Personal Use Monitors used to characterize puff topography in natural environment studies of tobacco and electronic nicotine delivery system users. My work supports federally funded observation studies, including OS8, and involves commissioning monitors, maintaining calibration integrity, managing large scale deployment datasets, and conducting post deployment performance analysis.

I am actively involved in compiling and analyzing longitudinal calibration data, assessing monitor reliability using statistical metrics such as intraclass correlation coefficients, and evaluating sensor stability across repeated laboratory and field use. I also support emissions testing and product characterization studies and am currently leading the preparation of a manuscript on the validation and performance evaluation of third generation wPUM monitors.

Publications and Manuscripts in Preparation

Mbaka, C. G., Hensel, E. C., Robinson, R. J. Assessing Monitors for Characterizing the Natural Environment Puff Topography of Tobacco Product Users. Manuscript in preparation, June 2025.

Technical Skills

Embedded systems development, sensor based measurement systems, data analysis, and computational modeling.
Programming experience includes C, Embedded C, Python, MATLAB, and Assembly.
Platforms and tools include STM32 microcontrollers, ARM Cortex M architectures, FreeRTOS, CMSIS, MATLAB, Simulink, Git, CubeMX, Keil uVision, and Visual Studio Code.
Experience with real time systems, signal processing, control systems, and experimental instrumentation.

Prior Professional Experience

Weapon Electrical Officer, Lieutenant Commander
Nigerian Navy
April 2012 to September 2023

I supervised the installation, maintenance, and operational readiness of electrical, electronic, and computer systems across naval platforms. My responsibilities included ensuring system reliability during missions and overseeing complex technical operations. My final appointment was as Staff Officer 2 at the Nigerian Naval Headquarters prior to retirement.

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List of Acronyms and Nomenclature

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1 SPECIFIC AIMS

1.1 FIELD OF STUDY

This work is situated in the field concerned with understanding how real-world use of inhalant tobacco products relates to measurable biological response in humans. A central challenge in this field is linking product characteristics and behaviour patterns of use to internal biomarkers that reflect tobacco exposure and tobacco-related health effects. While associations between product use and health outcomes are well established at a population level, quantitative frameworks that connect real-world use behavior to biological response at the individual level remain limited.

Product characteristics influence aerosol emissions, user behaviour and consumption while user behavior determines how products are consumed over time (Figure 1). Puff topography captures this behavior by describing the timing and intensity of use. Together, emissions and consumption determine how much material is delivered to the user during product use. Although these components can be measured independently, they are often treated separately in existing studies, limiting their usefulness for downstream biological interpretation.

In this work, behavior-based yield (BBY) is used as the exposure construct and is implemented as a model that combines consumption measures, such as daily puff count or daily puff volume, with emission characteristics to estimate the yield delivered to the user's mouth. BBY provides a quantitative link between real-world product use and biological analysis; however, approaches that rigorously validate BBY-derived exposure against biological measurements in human subjects are still underdeveloped.

A key gap in the field is the lack of integrated frameworks that translate BBY-based exposure into exposure biomarkers and subsequently evaluate how these responses relate to tobacco-related health effect biomarkers. Physics-based pharmacokinetic behavior-based yield (PKBBY) modeling offers a principled way to map BBY to exposure biomarkers, such as nicotine and cotinine, but has not yet been systematically demonstrated and validated using real-world ambulatory data. Furthermore, methods for classifying and interpreting PKBBY responses in relation to health-effect biomarkers and participant-level factors remain largely unexplored. Figure 1 illustrates the conceptual framework motivating this work, showing how product characteristics influence emissions, puff topography, and consumption behavior, how these processes are integrated into exposure through BBY, and how exposure may be related to exposure biomarkers and tobacco-related health effect biomarkers through PKBBY modeling and downstream analysis.

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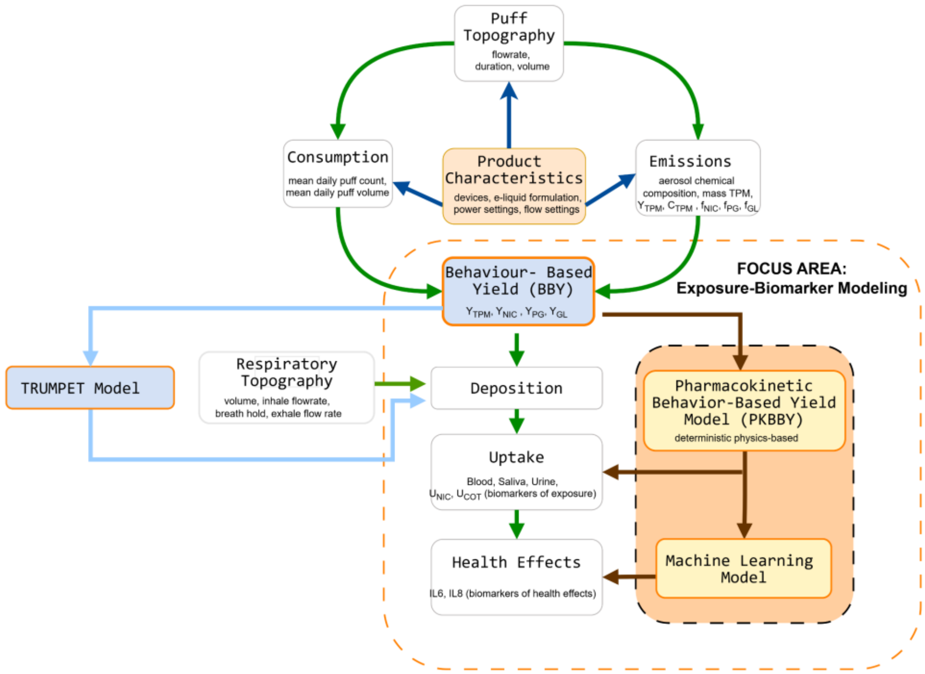

Figure 1: Conceptual framework linking product characteristics, emissions, puff topography, and consumption behavior to exposure through behavior-based yield and downstream biological response.

This dissertation addresses these gaps by developing and validating an integrated framework that combines real-world measurement, behavior-based exposure modeling, physics-based pharmacokinetic analysis, and data-driven classification. The focus of this work is not to establish clinical causality, but to determine whether exposure derived from real-world product use can be quantitatively linked to biological response in human participants.

1.2 Premise

Despite advances in measurement of real-world inhalant product use, quantitative exposure estimates derived from puff topography and emissions data are rarely validated against biological measurements in human subjects. Behavior-based yield provides a physically interpretable estimate of the yield delivered to the user's mouth, yet its ability to predict observed exposure biomarkers across individuals and usage conditions remains insufficiently demonstrated. As a result, uncertainty persists regarding how reliably exposure estimates derived from naturalistic use can be used to interpret biological response.

The premise of this dissertation is that exposure estimates derived from behavior-based yield, when coupled with physics-based pharmacokinetic modeling, can be systematically evaluated and validated using human subject data. By relating BBY-derived exposure to nicotine and cotinine measurements, and by examining downstream associations with tobacco-related health-effect biomarkers and participant-level factors, this work seeks to establish a defensible framework for interpreting real-world inhalant product use in terms of biological response.

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1.3 AIM 1. DEVELOP PROTOCOLS FOR AND CONDUCT VALIDATION, COMMISSIONING, AND PERFORMANCE ASSESSMENTS OF wPUM™ PUFF TOPOGRAPHY MONITORS.

Aim Description

The objective of this aim is to establish, document, and apply rigorous validation and performance assessment protocols for Wireless Personal Use Monitors (wPUM™) used to characterize puff topography in natural environment studies. Reliable exposure–health modeling depends on the accuracy, stability, and longitudinal performance of the measurement systems used to generate puff topography data. This aim focuses on ensuring that puff topography observations used in subsequent exposure and biomarker analyses are supported by validated and well-characterized instrumentation.

Hypothesis

The operational performance of ambulatory puff topography monitors changes over time as a function of repeated deployment, environmental exposure, monitor aging, and evolving sensing technologies. Systematic validation, commissioning, and longitudinal performance assessment methods can quantify these changes and improve confidence in puff topography measurements collected under natural environment conditions.

Approach

This aim develops and applies validation, commissioning, and operational assessment methods for successive generations of wPUM™ puff topography monitors using historical and contemporary datasets spanning OS4 through OS9. Calibration records, deployment summaries, commissioning data, and puff topography analyses are integrated to characterize monitor operating envelopes, longitudinal stability, deployment variability, and operational degradation.

Research Question 1.1 What is the complete validation process for a family of puff topography monitors (i.e., the wPUM Gen 4 Vuse Alto Monitor)?

Gap

A complete validation was completed for the Gen 2 wPUM puff topography monitor design and was published. Internal results have been compiled for almost all information needed to validate and publish the Gen 3 wPUM puff topography monitor design, but this work needs to be completed and published. Validation of the Gen 4 wPUM puff topography monitor design has not been started. While we know the RTL wPUM monitors are best-in-class, the actual performance of the monitors has not been fully documented in the literature.

Objective

Develop and document a complete validation protocol for next-generation wPUM puff topography monitors, including calibration procedures, operating-envelope characterization, puff-duration

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verification, volumetric assessment, repeatability analysis, and deployment-readiness evaluation.

Outcome

Publish a validation protocol (e.g., on protocols.io) with a persistent document identifier that can be referenced by future publications and promoted as a best-practice methodology for ambulatory puff topography monitor validation within the research community. This work will also support development of the Laboratory Study Protocol for FDA-related studies.

Research Question 1.2 How do the performance of wPUM Gen 3 monitors change over time from commissioning to retirement?

Gap

No publications have been found which document the performance history of puff topography monitors over time. Methods for quantifying performance must be articulated, results compiled, discussed, and published.

Objective

Quantify the operating envelope, accuracy, calibration agreement, deployment variability, and failure modes of Gen 3 wPUM puff topography monitors using historical and contemporary data collected during OS4 through OS8.

Outcome:

Publish a journal article documenting the operating envelope and lifetime performance history of multiple wPUM Gen 3 puff topography monitors using historical and contemporary data collected during OS4, OS5, OS6, OS7, and OS8. The work will identify and classify operational failure modes and performance degradation effects observed across repeated natural environment deployments.

Research Question 1.3 How do the performance of wPUM Gen 4 monitors change over time from commissioning to retirement?

Gap

When RQ 1.1 and 1.2 are addressed, protocols for assessing lifetime performance of a monitor fleet will have been defined and demonstrated. RQ 1.3 is focused on applying those methods to even newer monitors, with anticipated improved performance. The novel aspect of this work will be to quantify changes in performance from Gen 2 to Gen 3 to Gen 4 in an objective manner.

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Objective

Apply validation, commissioning, and longitudinal performance assessment methods to Gen 4 wPUM puff topography monitors and characterize monitor behavior across commissioning, deployment, sustained operation, and continued technological evolution.

Outcome

Publish a journal article documenting the lifetime performance history of multiple wPUM Gen 4 puff topography monitors using contemporary data from OS8 and OS9. The work will identify and classify operational failure modes and performance degradation effects observed across repeated natural environment deployments.

1.4 AIM 2. DEVELOP, CONDUCT AND DOCUMENT PUFF TOPOGRAPHY ANALYSIS METHODS, RESULTS, AND DATA INTERPRETATION OF PUFF TOPOGRAPHY, BEHAVIOR BASED YIELD (BBY), AND SESSION TOPOGRAPHY TO QUANTIFY THE BEHAVIOR OF HUMAN SUBJECT STUDY PARTICIPANTS.

Aim Description

The objective of this aim is to derive, characterize, and evaluate quantitative exposure metrics from validated puff topography data collected during ambulatory use. Building on the validated and longitudinally characterized measurement systems established in Aim 1, this aim focuses on transforming raw puff topography observations into exposure metrics that capture variability in consumption, emissions, and temporal structure of use. These metrics are intended to serve as inputs for predictive exposure–health modeling.

Hypothesis

Ambulatory puff topography observations contain sufficient temporal and behavioral information to derive quantitative exposure metrics that characterize participant-specific inhalant product use under natural environment conditions. Integrating puff topography with emissions modeling and session analysis will enable exposure characterization that more accurately reflects real-world constituent intake than conventional machine-puffing approaches or aggregated behavioral summaries.

Approach

This aim develops analytical workflows for deriving puff topography metrics, behavior-based yield (BBY), and session topography using ambulatory datasets collected during OS8 and OS9. Puff-level observations, emissions models, deployment metadata, and quality-control assessments are integrated to characterize participant-specific exposure behavior across products, sessions, and observational study conditions.

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Research Question 2.1 Can we identify and quantify the factors influencing quality control assessment of ambulatory puff topography observations (during OS8 and OS9) using Gen 3 and Gen 4 wPUM puff topography monitors?

Gap

Although quality-control assessment of puff topography data has been performed operationally since OS1, the methods, metrics, and analytical workflows used for assessing natural environment puff topography quality have not been comprehensively documented or published. Recent work performed during OS8 improved the robustness and consistency of this process, but these methods require formal definition, evaluation, and dissemination.

Objective

Develop and document analytical methods for assessing the quality and operational integrity of ambulatory puff topography datasets using calibration agreement, deployment summaries, participant observations, and puff topography consistency measures.

Outcome

Publish a journal article describing methods, metrics, results and analysis demonstrating how to assess the quality of natural environment puff topography data (and possibly how that assessment can leverage other data streams such as study coordinator notes, biomarkers, EMA.

Research Question 2.2 Can we quantify time response of ambulatory puff topography and BBY (using ENDS Emission Models) for study participants in two field studies (OS8 and OS9) using Gen 3 and Gen 4 wPUM puff topography monitors?

Gap

While the laboratory has published numerous studies describing puff topography behavior, fewer studies have investigated behavior-based yield using ambulatory observations integrated with emissions modeling. In addition, OS8 presents the first opportunity to quantify whether the presence of puff topography monitoring influences participant behavior as reflected by changes in salivary cotinine measurements collected under natural environment conditions.

Objectives

1. Develop methods for quantifying time-resolved puff topography behavior and behavior-based yield using ambulatory puff topography observations integrated with ENDS emissions models.
2. Evaluate whether the presence of ambulatory puff topography monitoring influences participant exposure behavior as reflected by changes in salivary cotinine measurements collected under natural-environment conditions.

OS8 Outcome

1. **Publish a journal article** describing the Emissions Model of the ENDS products used in OS8, including Geekbar Pulse (M and H), Pulse x (M and H), Hyde Edge, etc
2. **Publish a journal article** assessing the effect of topography monitoring on the cumulative evening salivary cotinine of subjects with and without topography monitoring.

OS9 Outcome

1. **Publish a journal article** describing the Emissions Model of the ENDS products used in OS9, including Vuse Alto, Golden Tobacco, 5% and 1.8% nicotine concentration.
2. **Publish a journal article** assessing the effect of nicotine concentration and ENDs experience level on statistical descriptions of puff topography behavior and cumulative daily exposure.

Results Documentation

1. **Table 1 Discrete Puff Table:** one topography record per puff per ppt, plus BBY results including Y_{TPM}, Y_{Nic}, Y_{PG}, Y_{GL}
2. **Table 2 Descriptive statistics by participant:** one record per ppt, averaged across all puffs, for all columns in Table 1, plus mean, standard deviation, count, CI95.

Research Question 2.3. Can we quantify session topography and BBY for study participants in two field studies

Gap

While puff-level topography behavior has been widely studied, fewer investigations have quantified session-level inhalant product use behavior and behavior-based yield across multiple ambulatory field studies conducted under natural environment conditions. Existing studies often focus on isolated puff descriptors or aggregated daily summaries without preserving session structure, cumulative daily exposure, or variability between observational study conditions. In addition, comparative analysis of session topography and BBY across studies involving different products, nicotine concentrations, participant populations, and study protocols remains limited.

Objective

Develop methods for quantifying and comparing session topography, cumulative daily exposure, and session-based behavior-based yield across two ambulatory field studies using validated puff topography observations collected during OS8 and OS9. These methods will characterize session structure, first-session-of-day behavior, cumulative daily exposure, interpuff timing patterns, and relationships between session behavior, nicotine concentration, product characteristics, and study-specific conditions.

Results Documentation

1. **Table 3:** one record per session for each Ppt
2. **Table 4:** one record for cumulative daily totals for each Ppt

OS8 Outcome (Observation Study)

Publish a journal article describing average session characteristics by participant as a function of study parameters (Nicotine Concentration and Product Characteristics).

OS9 Outcome (Clinical Study)

Publish a journal article describing average session characteristics by participant as a function of study parameters (Nicotine Concentration and ENDS Experience). Particular items of interest are daily average session stats, first session of the day stats (number of puffs, duration, flow rate and interval of each puff, and yield per first session), and time interval after the conclusion of the first session and the onset of the second session of the day.

1.5 AIM 3. DEMONSTRATE AND VALIDATE THE PHARMACOKINETIC BEHAVIOR BASED YIELD (PKBBY) MODEL FOR NICOTINE AND COTININE RESPONSE OF HUMAN SUBJECT STUDY PARTICIPANTS.

Aim Description

The objective of this aim is to demonstrate and validate a pharmacokinetic behavior-based yield (PKBBY) modeling framework that relates quantified exposure derived from real-world inhalant product use to biological response in human subjects. Building on validated puff topography measurements (Aim 1) and derived exposure metrics (Aim 2), this aim focuses on evaluating whether behavior-based yield can be used, in combination with pharmacokinetic modeling, to predict nicotine and cotinine response following product use.

Hypothesis

Participant-specific nicotine and cotinine biomarker response can be estimated using pharmacokinetic behavior-based yield (PKBBY) modeling derived from ambulatory puff topography observations, emissions-based exposure metrics, and first-order pharmacokinetic modeling. Integrating behavior-resolved exposure metrics with biomarker observations will enable quantitative characterization of participant-specific nicotine uptake and cotinine persistence under natural environment conditions.

Approach

This aim applies PKBBY modeling to ambulatory puff topography observations, emissions-derived exposure metrics, and biomarker measurements collected during OS8 and OS9. Participant-specific exposure metrics derived from Aim 2 are integrated with first-order nicotine and cotinine kinetic models to estimate nicotine uptake, cotinine formation, and biomarker persistence over time. Model validation is performed by comparing predicted biomarker dynamics with observed participant biomarker measurements collected during ambulatory and in-laboratory study conditions.

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Research Question 3.1 Can we estimate participant-specific time constants of cotinine, t_{cot}, decay using overnight patterns? Can we estimate t_{nic}, k_{nic}, and k_{cot} using ambulatory observations of biomarkers in conjunction with PKBBY model?

Gap

A preliminary feasibility study was previously published that showed promise. However, much more remains to be done. In particular, the PKBBY model needs to be refined and validated on a larger human subject study population.

Objective

Develop and validate methods for estimating participant-specific nicotine and cotinine pharmacokinetic parameters using overnight biomarker decay patterns and ambulatory puff topography observations integrated with PKBBY modeling.

Results Documentation

Table: Containing t_{cot}, t_{nic}, k_{nic}, and k_{cot} for each participant in OS8 and OS9 using over night decay and daily ambulatory exposure response

OS8 Outcome (Observation Study)

Publish a journal article validating the PKBBY model using human subject data

Research Question 3.2 Can we estimate the transient response of three in-lab vaping sessions using PKBBY model?

Gap

A validated PKBBY model may be able to demonstrate a causal relationship between tobacco product characteristics and biomarkers of exposure (e.g. cotinine) and health effect (e.g., IL6 or IL8).

Objective

Develop methods for applying participant-specific PKBBY models derived from ambulatory observations to transient nicotine and cotinine response during controlled in-laboratory vaping sessions.

Result Documentations

Time series showing coupled first order reaction kinetics of nicotine and cotinine overlaid with discrete biomarker responses and error quantification thereof.

OS9 Outcome (Clinical Study)

Publish a journal article using the validated PKBBY model to assess ambulatory (natural environment) response of participants to identify t_{cot}, t_{nic}, k_{nic}, and k_{cot} and then apply the participant-specific PKBBY model to three in-lab PK studies of nicotine and cotinine response.

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Research Question 3.3 Can we identify correlations between Ppt PKBBY response and study factors (*i.e.,* Nicotine concentration and ENDS experience) and confounding factors (*e.g.,* age, sex, BMI) using methods of machine learning, AI or similar techniques?

Gap

This is a stretch goal to explore what additional knowledge and insight we can harvest from the rich data sets spanning at least OS8 and OS9 and potentially harvesting archival data from OS1 through OS7.

Objective

Develop exploratory statistical learning and machine learning methods for identifying relationships between PKBBY response, participant characteristics, exposure behavior, and study factors using ambulatory observational datasets.

Results Documentation

Classification using interpretable and modern machine learning frameworks, including gradient-boosted decision tree models (XGBoost/LightGBM), k-nearest neighbor (KNN), clustering analysis, and exploratory deep learning architectures such as TabNet, GRU-attention networks, or Temporal Fusion Transformer (TFT) models for time-dependent PKBBY and exposure-response pattern analysis.

Outcome:

Present a conference poster or podium presentation or journal article documenting findings at a professional conference or in a journal of interest.

2 Significance And Literature Review

i. Tobacco-Related Disease and Public Health

Tobacco use is one of the leading causes of preventable disease and premature mortality worldwide [1]. Globally, tobacco consumption is responsible for more than eight million deaths each year, with the burden distributed across cancer, cardiovascular disease, respiratory disease, and other chronic conditions[1]. In the United States and internationally, tobacco smoking has been identified as the single most important avoidable risk factor contributing to morbidity and mortality at the population level [2]. The causal relationship between cigarette smoking and lung cancer was first established through epidemiological investigations conducted in the mid-twentieth century. Early case–control studies demonstrated a strong association between smoking intensity and bronchogenic carcinoma, providing quantitative evidence that tobacco smoke exposure substantially increased cancer risk [3]. These findings were corroborated by large prospective cohort studies, including the British Doctors Study, which showed significantly higher mortality among smokers compared to nonsmokers and demonstrated a dose–response relationship between smoking and disease outcomes [4].

Subsequent follow-up studies further confirmed that cigarette smoking increases mortality from lung cancer, cardiovascular disease, and chronic respiratory disease [5]. This body of evidence culminated in the 1964 United States Surgeon General's Report on Smoking and Health, which concluded that cigarette smoking is a cause of lung cancer and chronic bronchitis and is associated with increased mortality from coronary heart disease [6]. Since that time, successive Surgeon General reports and international assessments have expanded the list of smoking-related diseases, identifying causal links between tobacco smoking and cancers of the larynx, oral cavity, esophagus, pancreas, and bladder, as well as chronic obstructive pulmonary disease and stroke [2], [7]. These conclusions are supported by mechanistic, clinical, and population-level evidence demonstrating that repeated inhalation of tobacco smoke produces systemic biological effects that accumulate over time [8]. Despite decades of public health intervention and declining smoking prevalence in some regions, tobacco use remains a major global health problem. Population-level declines in combustible cigarette smoking have been accompanied by the emergence and rapid adoption of alternative inhaled nicotine products, including electronic nicotine delivery systems and tobacco heating products [9]. These products differ from conventional cigarettes in their design, aerosol generation mechanisms, and patterns of use, yet they still deliver nicotine and other constituents capable of producing biological effects [10]. The long-term health consequences of chronic use of these products remain uncertain, particularly as device designs, consumable formulations, and user populations continue to evolve [9].

A central challenge for public health is understanding how inhalant tobacco product use translates into biological response that may underlie disease risk. Health outcomes associated with tobacco use do not arise solely from the presence of a product, but from repeated exposure driven by how the product is used over time and how resulting dose interacts with biological systems [11]. Biomarkers such as nicotine and its primary metabolite cotinine are widely used indicators of tobacco exposure and are routinely measured in clinical and population studies[8]. However, interpretation of these biomarkers depends on understanding the exposure context that produced them, including variability in consumption behavior, product characteristics, and temporal patterns of use [8], [12]. Without quantitative frameworks that connect real-world inhalant product use to measurable biological response, assessments of tobacco-related disease risk remain incomplete. Many population studies rely on categorical measures of product use, such as current versus former use, which do not capture variability in intensity, frequency, or temporal structure of consumption[9]. This limitation constrains the ability to evaluate emerging products, compare risk across product categories, and interpret biomarker data in the context of real-world exposure. Addressing tobacco-related disease therefore requires approaches that link patterns of inhalant product use to biological response using exposure constructs that reflect how products are actually consumed. This public health need provides the foundation for the subsequent sections of this dissertation, which focus on measurement, exposure characterization, and biological interpretation.

ii. Characteristics of Inhalant Products and Puff Topography Monitors

Inhalant tobacco products exhibit substantial diversity in physical design, operating principles, and consumable formulations, all of which influence aerosol generation and user interaction.

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Combustible cigarettes, electronic nicotine delivery systems (ENDS), and tobacco heating products differ in flow path geometry, resistance to draw, power delivery, aerosol formation mechanisms, and chemical composition of emissions, leading to distinct patterns of nicotine delivery and toxicant exposure [8], [13], [14]. Even within a single product category, variations in device design, consumable formulation, and operating conditions can produce large differences in aerosol emissions and user experience [13], [14].

Human consumption of inhalant products is adaptive rather than passive. Users modify puff flow rate, puff duration, interpuff interval, and session structure in response to product characteristics such as nicotine concentration, aerosol harshness, draw resistance, and sensory feedback [15], [12]. These compensatory behaviors directly violate the assumptions underlying standardized machine puffing protocols, which apply fixed puff volumes, durations, and intervals that do not reflect real-world use[16],[17], [18]. As a result, emissions measured under machine conditions often fail to represent the material actually delivered to users during real world consumption.

Puff topography provides a quantitative description of the time-varying structure of inhalant product use, including puff flow rate, duration, volume, and daily consumption patterns. Extensive prior work has shown that puff topography varies widely across individuals, products, and environments, and that this variability is a primary determinant of delivered yield and exposure [19], [20]. Accurate characterization of puff topography is therefore essential for interpreting how inhalant products are used outside the laboratory and for linking product characteristics and emissions to downstream exposure and biomarker measurements.

Early puff topography studies relied on laboratory observation, self-reported behavior, or tethered measurement systems that constrained mobility and altered natural use patterns [21], [22]). Subsequent development of portable monitoring devices enabled more realistic measurement of user behavior, but early systems were limited in resolution, storage capacity, deployment duration, or long-term stability [23], [24], [25]. These limitations restricted the ability to capture extended consumption patterns and day-to-day variability under real-world conditions.

Wireless Personal Use Monitors (wPUM™) were developed to address these challenges by enabling sustained ambulatory measurement of puff topography across multiple inhalant product categories. Prior work demonstrated that wPUM monitors can capture high-resolution puff flow, duration, volume, and daily consumption under natural environment conditions, revealing behavior patterns that differ substantially from laboratory protocols [17], [24], [26]. These findings established that exposure cannot be inferred from product characteristics or emissions testing alone and that behavior-resolved measurement is required to characterize real-world use.

The ability to quantify puff behavior under natural environment conditions is particularly important because inhalant product exposure is shaped not only by product design and aerosol emissions, but also by how products are consumed over time. Variability in puff flow rate, puff duration, interpuff interval, and daily consumption patterns directly influences aerosol delivery and nicotine exposure. Accurate ambulatory puff topography measurement is therefore essential for characterizing real-world inhalant product use and for supporting quantitative assessment of exposure.

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iii. Significance of Behavior-Based Yield and Exposure–Biomarker Modeling

A central limitation in inhalant tobacco research is the absence of exposure metrics that accurately represent how products are consumed under real-world conditions and how that use translates into internal biological dose. Traditional tobacco exposure assessment has relied heavily on machine-generated yields produced under standardized machine-smoking protocols, including ISO and FTC regimens. These protocols apply fixed puff volumes, durations, and interpuff intervals that do not reflect the adaptive nature of human puffing behavior [16], [19]. Extensive prior work has demonstrated that tobacco users modify puffing behavior in response to nicotine delivery, draw resistance, sensory feedback, and product design characteristics, resulting in exposure patterns that differ substantially from machine predictions [21], [20].

The limitations of machine-derived exposure estimates became particularly evident during the introduction of "low-tar" and "light" cigarettes. Although these products produced lower nicotine and tar yields under standardized machine-smoking conditions, reductions in population-level disease risk did not occur because smokers compensated behaviorally by increasing puff volume, puff duration, puff frequency, or total cigarette consumption [27], [28], [8]. Similar concerns have emerged with modern electronic nicotine delivery systems (ENDS), where differences in device power, aerosol generation, nicotine formulation, and user behavior produce highly variable exposure profiles that cannot be inferred from nicotine concentration or emissions testing alone [13], [14]. These findings demonstrate that exposure cannot be interpreted solely from product emissions and that consumption behavior must be incorporated directly into exposure assessment.

Behavior-based yield (BBY) was introduced to address this limitation by explicitly integrating emissions characteristics with measured puff topography behavior. BBY represents the constituent yield delivered to the user as a function of both emissions per unit aerosol and the temporal structure of puffing behavior, including puff flow rate, puff duration, puff volume, puff timing, and puff frequency [29]. By construction, BBY preserves compensatory behavior and captures how users adapt to different product designs and nicotine formulations during real-world consumption. If a user compensates for reduced nicotine concentration by increasing puff intensity or consumption frequency, the resulting BBY correspondingly increases. Prior work demonstrated that BBY varies substantially across users and products and that products marketed with lower nicotine concentration do not necessarily produce lower delivered nicotine once user behavior is considered[29], [30]. Consequently, BBY provides a behavior-resolved exposure metric that more accurately reflects real-world intake than traditional machine-derived yields.

Despite these advances, BBY remains underutilized in exposure–response research, and relatively few studies have established frameworks for deriving, validating, and interpreting BBY metrics from extended ambulatory puff topography datasets. Prior ambulatory puff topography studies successfully characterized real-world consumption behavior across users and products [24], [31], while separate emissions studies quantified constituent yields under laboratory conditions [32], [13], [33]. However, relatively few frameworks have integrated emissions characterization with time-resolved puff behavior to estimate constituent intake under natural environment conditions [29]. As a result, exposure assessments frequently obscure variability in cumulative intake, session structure, consumption rate, and temporal use patterns that may be important for biological interpretation.

An important limitation of BBY is that it remains an external exposure metric and does not directly describe biological uptake or internal dose. Biomarkers such as nicotine and cotinine reflect the combined effects of inhaled yield, deposition, absorption, distribution, metabolism, and elimination, processes that evolve dynamically over time and vary substantially across individuals [34], [35], [36]. Existing pharmacokinetic studies have primarily relied on controlled dosing conditions and simplified representations of nicotine intake that do not fully capture the temporal variability and behavioral complexity associated with real-world inhalant product use [37], [38], [39]. Consequently, a substantial gap remains between behavior-resolved exposure estimates and measured biomarkers of exposure in human subjects.

Pharmacokinetic behavior-based yield (PKBBY) modeling provides a framework for bridging this gap by treating BBY as a time-resolved input function to pharmacokinetic models. Within this framework, nicotine yield derived from ambulatory puff topography and emissions modeling is propagated through first-order pharmacokinetic processes to estimate nicotine and cotinine biomarker dynamics over time[39]. Preliminary feasibility analyses demonstrated that PKBBY modeling can reproduce observed cotinine dynamics and estimate participant-specific pharmacokinetic behavior using ambulatory data; however, the framework has not yet been systematically refined or validated across larger study populations or extended deployment periods[39]. These limitations motivate the need for more comprehensive approaches capable of integrating validated puff topography measurement, exposure modeling, and biomarker analysis into a unified framework.

A further limitation of the current literature is that exposure modeling and health-effect evaluation are frequently treated as separate domains. Nicotine and cotinine are well-established biomarkers of tobacco exposure [35],[36], while numerous biomarkers associated with tobacco-related disease processes have been reported in cigarette smokers and electronic cigarette users. These include inflammatory cytokines such as Interleukin-6 (IL-6) , Interleukin-8 (IL-8) , and Tumor Necrosis Factor-alpha (TNF-α), oxidative stress biomarkers, cardiovascular injury biomarkers, immune dysregulation markers, and respiratory injury biomarkers [40], [41], [42], [43],[44], [45], [46], [47], [48]. However, these biomarkers are rarely analyzed in conjunction with behavior-resolved exposure metrics derived from ambulatory puff topography and constituent exposure modeling [29], [39].

This dissertation addresses these limitations by advancing an integrated framework that combines validated ambulatory puff topography measurement, behavior-based exposure modeling, pharmacokinetic analysis, and biomarker interpretation. Within this framework, BBY serves as the quantitative exposure construct linking product emissions and consumption behavior to internal biological response. PKBBY modeling uses behavior-resolved nicotine intake derived from real-world ambulatory puff topography to estimate nicotine and cotinine biomarker dynamics through first-order pharmacokinetic processes. Downstream analyses examine relationships between PKBBY-derived exposure profiles, inflammatory biomarker response, and

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participant-specific behavioral and physiological characteristics using statistical and data-driven modeling approaches. By integrating validated puff topography measurement with behavior-based exposure modeling and biomarker analysis, this work establishes a unified framework for relating real-world inhalant product use to biological response under natural environment conditions.

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3 Innovation

3.1 Aim 1

3.2 Aim 2

3.3 Aim 3

4 Methods and Materials

4.1 Aim 1

4.2 Aim 2

4.3 Aim 3

5 Preliminary Results

5.1 Aim 1

5.2 Aim 2

5.3 Aim 3

6 Discussion

6.1 Discussion of Preliminary Results

6.2 Dissemination Plan

6.3 Broader Impacts of the Proposed Work

7 Conclusions and Future Work

7.1 Preliminary Findings

7.2 Future Work (Beyond the Scope of the Dissertation)

7.3 TIMELINE

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